Application of Modern Tabular Deep Learning Architectures for ADME Prediction based on structural SMILES Representation

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Abstract

Accurate prediction of ADME (Absorption, Distribution, Metabolism, and Excretion) properties is crucial in drug discovery. Recent advances in tabular deep learning models including Feature Tokenizer Transformers (FTT), Neural Decision Forests (NDF), and Tabular Prior-Data Fitted Network (TabPFN) claim to outperform traditional methods like eXtreme Gradient Boosting (XGBoost) and Multilayer Perceptrons (MLPs) on tabular data. This study systematically evaluates these modern architectures against established machine learning techniques in the domain of chemoinformatics using tabular data for ADME prediction. Using publicly available Biogen datasets, we benchmark various models on six key ADME endpoints. Our results show that while modern deep learning models perform comparably, they do not consistently surpass traditional methods.

Introduction

Accurate prediction of ADME properties is an important step in drug discovery and development that influences the safety and efficacy of a compound.[1] Poor ADME properties can cause late-stage drug failures, making early computational screening essential to reduce costs and improve success rates in pharmaceutical research.[2,3] Traditional in vitro and in vivo ADME assays are time-consuming and expensive, hightlighting in silico approachs with the use of machine learning (ML) to predict these properties using molecular descriptors and structural fingerprints.[4,5]

During the past two decades, ML techniques such as Random Forests,[6] Support Vector Machines,[7] and Gradient Boosting[8] have demonstrated progress in successful ADME prediction tasks.[9–11] XGBoost[12] has been widely adopted due to its outperforming results over other ML models on tabular data.[13,14] Deep learning approaches are gaining more and more traction in cheminformatics with models such as Multilayer Perceptrons (MLPs) and graph neural networks (GNNs) being applied to ADME prediction.[4,15,16] However, deep learning models often require large datasets, extensive computational resources, and careful hyperparameter tuning to outperform competitively against traditional ML methods on tabular datasets.[17]

This necessity for large sample sizes and the challenges associated with pretraining and fine-tuning make representation learning less practical in many cheminformatics scenarios. As a result, it is often more convenient and effective to leverage expert-based representations, which can provide robust performance without the extensive resource demands of deep learning models.

Since tabular data is the most common data type for real-world ML applications, several deep learning architectures are specifically tailored for tabular data and have been proposed as potential improvements over traditional ML models.[17] These include the Feature Tokenizer Transformer (FTT),[18] which applies self-attention mechanisms to tabular feature embeddings; Neural Decision Forests (NDFs),[19] which integrate differentiable decision trees with deep neural networks; and TabPFN - a Transformer-based model designed for rapid tabular data inference using pre-trained Bayesian priors.[20] Proponents of these architectures argue that these new architectures offer superior performance in tabular prediction tasks over conventional ML algorithms for tabular data. Despite the proven advantages of these modern architectures, their practical benefits over established techniques remain unclear in the domain of chemoinformatics.

In this study, we systematically compare modern tabular deep learning architectures like FTT, NDF and TabPFN against conventional machine learning methods including XGBoost and MLPs on tabular ADME data. Using publicly available Biogen datasets, we evaluated model performance across six key ADME endpoints to determine whether modern tabular-based deep learning approaches offer a tangible advantage. Our findings suggest that while those modern architectures can achieve competitive results, they do not consistently outperform well-established ML techniques.

Methods

Data

Fang *et al.*[4] collected ADME properties for a set of test compounds over a 20-month period covering six endpoints: human plasma protein binding (hPPB), rat plamsa protein binding (rPPB), Solubility (Sol), human liver microsomal stability (HLM), rat liver micrisomal stability (RLM) and MDR1-MDCK efflux ratio (MDR1-ER). After each time split, machine learning models were trained on the previous collected data to predict ADME activity for compounds in the subsequent period. The study adhered to standardized experimental protocols and preprocessing steps to ensure high-quality measurements. Biogen also published compounds from commercially available libraries such as Enamine and eMolecules spanning a chemical space comparable to that of the time-split experiment. Table 1 provides an overview of the number of compounds per endpoint and the distribution of data between training and test sets. To standardize the activity values across endpoints, all measurements are log-transformed using base 10.

Table 1: Min, Median and Max log activity value of 3521 unique compounds from the public Biogen compoundset distributed per endpoint and train/test-split

Endpoint	Dataset	min	median	max	count
HLM	Test	0.676	1.282	3.373	618
	Train	0.000	1.183	3.340	2469
MDR1_ER	Test	-1.162	0.116	2.282	529
	Train	-1.047	0.156	2.725	2113
RLM	Test	0.000	2.274	3.786	611
	Train	0.000	2.321	3.970	2443
Sol	Test	-1.000	1.521	2.136	435
	Train	-1.000	1.546	2.179	1738
hPPB	Test	-1.155	0.736	2.000	362
	Train	-1.593	0.660	2.000	1446
rPPB	Test	-1.638	0.876	2.000	177
	Train	-1.403	0.782	2.000	708

Structure Standardizing

We standardize the SMILES Strings with RDKIT (version 2023.9.4) which involves the following steps: Remove of H Atoms, disconnect metal atoms, normalize molecules, reionize molecules, selecting the "parent" fragment if multiple fragments are available, neutralize molecules and canonicalizing tautomers.

Molecular Representations

We use the molecular representations from Fang *et al.*[4] to ensure a fair comparison to their public benchmark. This feature set consists of 1340 features containing 1024 functional connectivity fingerprint descriptors (FCFP4 with a radius of 4 and 1024 bits) and 316 2D RDKit[21] descriptors. Furthermore, the 316 2D RDKit descriptors are scaled with a StandardScaler to prevent stability problems while training.

Machine Learning Methods

Extreme Gradient Boosting

Extreme Gradient Boosting (XGB)[12] is a machine learning technique based on the concept of gradient boosting[22,23] on decision trees[24,25] enhanced with regularization and optimization techniques such as Lasso/Ridge regularization or learning rate shrinkage. The algorithm builds up an ensemble of small decision trees through boosting. Those trees are then trained in such a way that each tree minimizes the prediction errors of the previous tree. XGB works especially well on tabular data and even outperform Neural Networks becoming SOTA regarding tabular data.[13,17] Equation 1 shows the loss function with regularization of XGBoost where l is a function of CART learners,[25] f_t a function to improve the model and Ω a regularization term for the model complexity.[12] For those reasons, there have been many successfull applications of XGB in the domain of chemoinformatics such as ADME prediction tasks[4,5] or quantitative structure–activity relationship (QSAR) modeling.[14] We use the Python package "xgboost" version 2.0.3 for our ADME prediction study.

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l(y_i, \hat{y}_i^{t-1} + f_t(x_i)) + \Omega(f_t) \tag{1}$$

Multilayer Perceptron

A Multilayer Perceptron (MLP) (also known as Artifical Neural Network or Fully Connected Neural Network) consists of an input layer, at least one hidden layer and an output layer. Each layer is composed of neurons that are fully connected to the neurons in the adjacent layers. The output of the input and hidden layers are transformed by a nonlinear activation function, allowing the model to learn complext patterns between the data and the target through. Equation 2 outlines the propagation through one layer-block of a MLP. The prediction output is used to calculate a loss to adapt the parameters of the MLP with the backpropagation algorithm.[26] MLPs are already used for QSAR-Modelling[15] or ADME-Prediction.[27] We use dropout layers[28] layers to reduce risk of overfitting. Those dropout layers are later used for Monte-Carlo Dropout[29] to generate predictions on the test set by building an ensemble of different MLP models through toggeling of randomly selected neurons. The MLPs are build with the open-source software PyTorch version 2.1.1.[30]

$$MLP(X) = Act(Drop(Lin(X))) \tag{2}$$

Feature Tokenizer Transformer

Feature Tokenizer Transformer (FTT)[18] is a Transformer[31] based neural network architecture designed for tabular data. The architecture consists of a feature tokenizer to embed numerical and categorical features into learnable vector representations. We project the input fingerprints and descriptors to 512 features with a linear layer to save GPU Memory since the proposed featurizing of Fang *et al.*[4] consists of 1340 features. We also modify the FTT to avoid using categorical features, since the cheminformatic descriptors only contains numerical features. A numerical embedding is learned through the linear projection of a feature x_j to an embedding T_j (Equation 3). A [CLS] token is introduced to aggregate information across features (Equation 4). Those embeddings are passed through a stack of Transformer layers where each Transformer layer consists of multi-head self-attention and feedforward layers with residual connections and layer normalization. The final representation of the [CLS] token is used for prediction in combination with a small MLP 5. We use the FT-transformer package version 0.3.0 in combination with PyTorch version 2.1.1.

$$T_j = b_j + x_j \cdot W_j \in R^d \tag{3}$$

$$T = stack[[CLS], T_1, ..., T_m] \in R^{m+1 \times d} \tag{4}$$

$$\hat{y} = Linear(Act(LayerNorm(T_L^{[CLS]}))) \tag{5}$$

Neural Decision Forests

Neural Decision Forests (NDF)[19] integrate decision trees with deep neural networks. NDFs replace the final layers with decision trees that learn feature representations and classifications jointly. They introduce stochastic and differentiable decision nodes, allowing gradient-based optimization via backpropagation. Equation 6 shows the routing function μ_l denoting the probability that a sample x will traverse to the output-leaf l. The index $n \in N$ corresponds to all internal nodes where each node has a parametrized decision function d_n based on a Bernoulli random variable. $\bar{d}$ refers to $1 - d_n$. The indicator function 1 is conditioned on the binary relation $l \swarrow n$ or $n \searrow l$ indicating if l belongs to the left or right subtree of node n. We adapt the NDF algorithm by switching out the sigmoid function with the identity function making NDFs suitable for regression tasks. We use the NDF version 0.3.0 in combination with PyTorch version 2.1.1.

$$\mu_l(x|\Theta) = \Pi_{n \in N} d_n(x;\Theta)^{1_{l \swarrow n}} \bar{d}_n(x;\Theta)^{1_{n \searrow l}} \tag{6}$$

TabPFN

Tabular Prior-Data Fitted Network (TabPFN)[20] is a Transformer[31] model designed for small tabular tasks. It performs in-context learning (ICL) by leveraging pre-trained knowledge to make predictions in a single forward pass. TabPFN is pre-trained on synthetic tabular datasets generated using a Bayesian prior incorporating causal reasoning[32,33] and structural causal models (SCMs).[34,35] The training process uses a Transformer-based PFN that approximates Bayesian inference over a large set of simulated data-generating mechanisms. The input data is transformed into feature tokens. A trained TabPFN model processes input features using attention-based mechanisms. Predictions are derived from a learned Posterior

Predictive Distribution (PPD) in a single inference step. This can be seen in Equation 7: $P(y|X)$ is the posterior predictive distribution of the output y given the input features X and model parameters θ. $P(\theta|X)$ is the posterior distribution of the parameters given the input data. We use TabPFN version 2.0.2.

$$P(y|X) = \int P(y|\theta)P(\theta|X)d\theta \tag{7}$$

In Vitro ADME Assays

In the following the in vitro ADME assays from Fang *et al.*[4] are described along the most important steps.

Plasma Protein Binding (hPPB & rPPB)

Plasma protein binding was evaluated using rat and human plasma collected in potassium EDTA (BIOIVT, Westbury, NY) with a rapid equilibrium dialysis (RED) device (ThermoFisher Scientific, Waltham, MA). The compounds were prepared at a concentration of 1 μM and spiked into pooled plasma samples from at least 10 donors. A volume of 300 μL of spiked plasma was added to the donor chambers of the RED device while 500 μL of phosphate buffer (100 mM potassium phosphate, 150 mM NaCl, pH 7.4) was added to the corresponding receiver chambers. The samples were incubated at 37 degree celsius with 5% carbon dioxide for 4 hours. 50 μL aliquots were collected from both donor and receiver compartments and transferred to a 96-well plate containing 200 μL of a protein precipitation solution (50:50 acetonitrile/methanol) with an internal standard. Plasma and buffer samples were vortexed for 30 seconds and centrifuged at 3900 rpm for 10 minutes. 50 μL of the supernatant was transferred to a clean 96-well plate containing 200 μL of 0.1% formic acid in a 90:10 water/acetonitrile mixture. The samples were analyzed using LC-MS/MS with a high-throughput injection system.

$$\%\text{Unbound} = \frac{\text{avgPAR in the buffer side}}{\text{avgPAR in the plasma side}} \times 100 \tag{8}$$

Solubility

Solubility was assessed using Analiza's patented automated miniaturized gold-standard shake flask method resulting in total equimolar chemiluminescent nitrogen detection to determine compound solubility in phosphate-buffered saline (PBS) at pH 6.8, with a final dimethyl sulfoxide (DMSO) concentration of 2%.

Liver Microsomal Stability (RLM & HLM)

Human liver microsomes (HLM) and rat liver microsomes (RLM) were used to assess compound metabolism. Compounds were incubated at a final concentration of 1 μM with liver microsomes (0.5 mg/mL) in a 0.1 M sodium phosphate buffer (pH 7.4) containing 3.3 mM magnesium chloride and 1 mM nicotinamide adenine dinucleotide phosphate (NADPH) at 37 °C. At specific time points (0, 5, 10, 15, 25, and 40 minutes) after NADPH addition, 40 μL aliquots were collected and transferred into wells of a 96-well plate containing 50 ng/mL of 8-cyclopentyl-1,3-dipropylxanthine (CPDPX, internal standard) in 200 μL of an acetonitrile:methanol mixture (1:1, v/v). The plates were then centrifuged at 3220g for 10 minutes. 40 μL of the supernatant from each well was transferred to a separate 96-well analytical plate containing 180 μL of a water:formic acid solution (100:0.1, v/v). The samples were thoroughly mixed and directly injected into an LC/MS/MS system for analysis. The intrinsic clearance (CLint) was determined using a standard equation for microsomal stability:

$$CL_{int} = \frac{0.693}{T_{1/2}} \times \frac{\text{incubation volume}}{\text{mg of microsomal protein}} \tag{9}$$

MDR1-MDCK Efflux Ratio

The assay utilized human MDR1-transfected MDCK cells (NIH cell line licensed from Absorption Systems (PA, USA)) to evaluate compound permeability and efflux. Compounds were prepared at a concentration of 1 μM in transport buffer (Hank's balanced salt solution with HEPES). MDR1-MDCK cells were cultured for seven days in 96-well Transwell insert plates (Corning, NY, USA). Before starting the assay, the inserts were washed and transepithelial electrical resistance (TEER) was measured.

85 μL of test compound solution was added to the donor compartment for apical-to-basolateral (A-B) transport, while 260 μL was added for basolateral-to-apical (B-A) transport. The receiver compartment contained transport buffer supplemented with 1% BSA with volumes of 250 μL for A-B and 75 μL for B-A transport. A 10 μL sample was taken from the donor compartment at time zero (T = 0).

The assay plates were incubated for 120 minutes. At the endpoint (T = 120 min), 10 μL and 50 μL samples were collected from the donor and receiver compartments. Donor samples were diluted with 40 μL of transport buffer containing BSA and a crash solution (110 μL of acetonitrile with an internal standard) was added to all samples. After centrifugation, 50 μL of the supernatant was transferred to a clean plate and mixed with 50 μL of water. The samples were analyzed using LC-MS/MS with a high-throughput injection system.

Analyte/internal standard area ratios were used to calculate apparent permeability (Papp), efflux ratio and mass recovery. These values were determined using standard equations based on the cumulative concentration in the receiver compartment over time:

$$P_{app} = (\frac{dC_r}{dt}) \times \frac{V_r}{(A \times C_E)} \tag{10}$$

$$\text{Mass balance} = 100 \times \frac{((V_r \times C_r^{\text{final}}) + (V_d \times C_d^{\text{final}}))}{(V_d \times C_E)} \tag{11}$$

Where $\frac{dC_r}{dt}$ is the cumulative concentration in the receiver compartment, V_d is the volume of the donor compartment, A the area of the inset, C_e the estimated experimental concentration, C_r^{final} the concentration of the receiver at the end of the incubation period and C_d^{final} the concentration of the donor at the end of incubation period.

Hyperparameter Search

We use the Optuna hyperparameter optimization framework,[36] using the Tree-structured Parzen Estimator (TPE)[37] algorithm. The evaluation loop aims to maximize R2 score in 5 fold-cross-validation. We use a Bayesian optimization algorithm to accelerate the hyperparameter search, given the computational cost associated to each evaluation.[38] Each model is tuned on each endpoint individually, with a budget of 100 trials. The only exceptions are TabPFN and NDF with 50 trials because those models have much smaller hyperparameter spaces. We also track the mean 5 fold cross-validation mean absolute error (MAE) since a good regression model is not soley defined by its R2 performance.[39] We use the model with with a R2 score which is greater equals the 0.99-R2 quantile value and MAE which is less equals the 0.01-MAE quantile value of all trials. If such a model does not exist the R2 quantile is gradually lowered while the MAE quantile is moderately set higher in small steps until a suitable model is found.

Figure 1 outlines the optuna workflow which is done for every endpoint-model combination. Each model architecture defines its own hyperparameter space from which the TPE-sampler chooses a hyperparameter configuration. This configuration is used to instanciate a model based on the configuration. A 5-fold-cross-validation is used to to validate the current hyperparameter configuration. For each fold a model instance is created, trained and evaluated on the validation fold. The R2 scores of the validation folds are aggregated as a mean value which is received by the sampler to find the hyperparameter configuration which maximizes the mean validation R2 score for each endpoint-model pair.

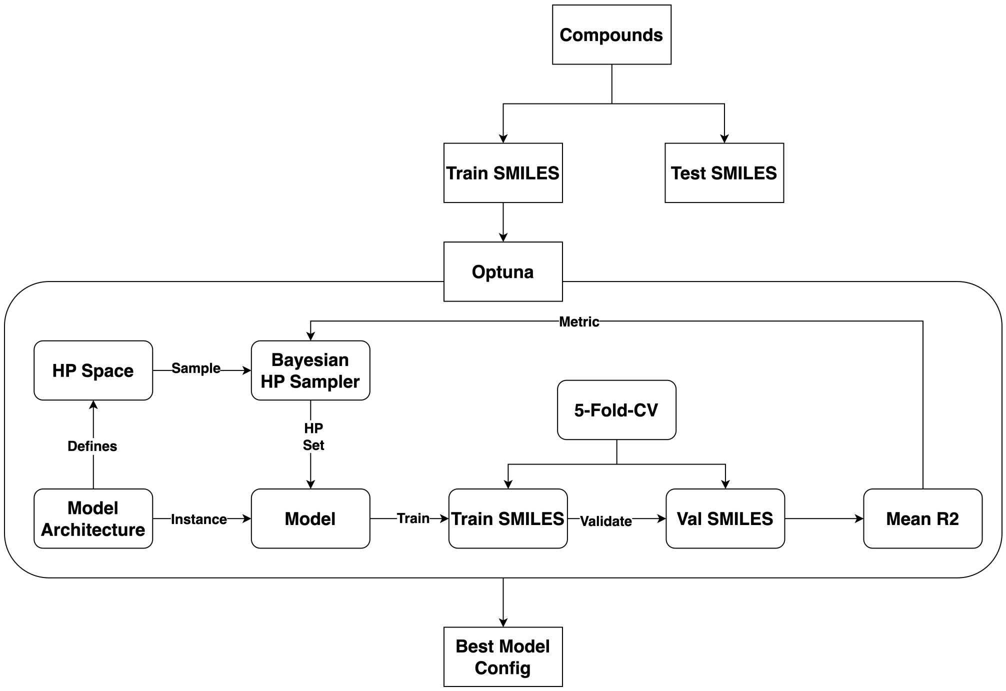


Figure 1: Schema of hyperparameter tuning with optuna for every endpoint-model combination resulting in a best hyperparameter configuration for each combination.

Evaluation

Each resulting endpoint-model combination of the hyperparameter tuning is retrained 30 times with different random seeds to generate a distribution of the test metrics on both test sets capturing the variance of random-dependent hyperparameters or weight-initializations.[18,20] To use the maximum amount of data, the training and validation data have been merged to build the final training set. After each training run, the trained model predicts on the unseen test data to calculate the test metrics: MAE, R2 and Spearman Rank Correlation (SPR). Inspired by the suggestions from Ash *et al.*[39] the resulting distributions of the test metrics are used for statistical methods like a repeated measures ANOVA and a Tukey honestly significant difference (HSD) post-hoc test to measure significant differences between the test metric distributions of the evaluated models. The Cohens D[40] measurement is used to measure the pairwise effect size as a difference between mean values of two distributions standardized with the mean standard deviation.

We also track the individual prediction scores of the best model of each model-training set combination for each testset. The mean of those prediction scores is used to investigate the applicability domain and the generalization ability of each model with the procedure from Fang *et al.*[4] The average similarity between a test compound and its five nearest neighbors in the training set is used to quantify the distance to the models domain of applicability. Structural similarity is measured using the Sorensen-Dice coefficient based on FCFP4 fingerprints[41] (1024 bits and a radius of 2). The resulting similarity scores were binned into intervals of 0.1 units. The Prediction error was defined as the absolute difference between experimental and predicted values for each individual test compound transformed by the decadal logarithm. We performed this analysis for each endpoint-model-trainingset combination.

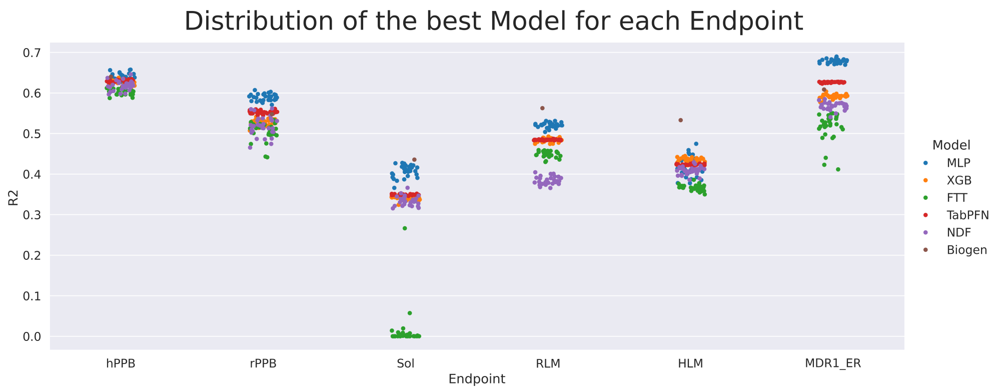
Results

MLP achieves superior Test R2 and MAE scores for most of the Endpoints

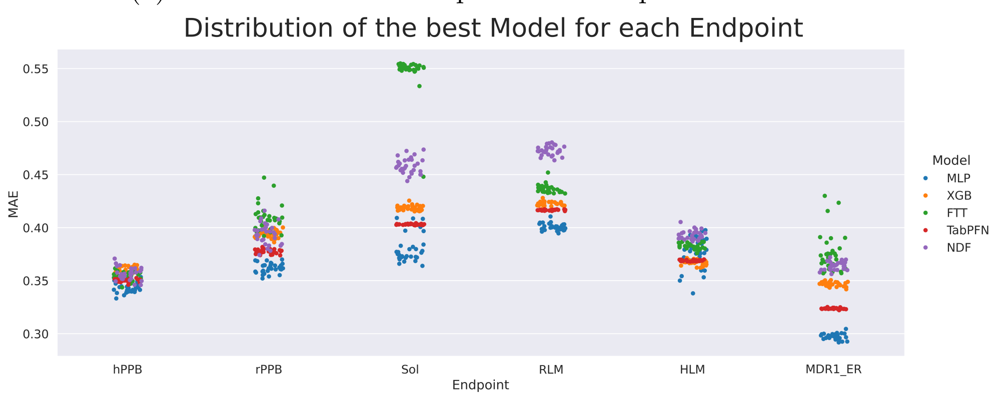
We first evaluate the test R2 (Figure 2a) and MAE (Figure 2b) values for each model-endpoint combination. Each point resulted from a training and evaluation process with a different random seed resulting in 30 different points per model-endpoint combination. The MLP shows the best performance among both metrics for every endpoint even outperforming Fang *et al.*[4] best models on MDR1-ER, hPPB and rPPB. XGB and TabPFN sharing the second best performances while the FTT and the NDF have the worst performances. This is in particular visible for the FTT R2 performance on solubility since most of the test metrics scatter around 0 indicating a model collapse. These rankings are more noticeable in terms of test MAE because the performance gaps are stronger than for the R2 metric.

Table 2 shows the mean and standard deviation values for R2, MAE and Spearman Correlation Coefficient (SPR) on the test sets. The bold entries hightlight the highest mean

R2 and lowest mean MAE values showing that the XGB model achieved the best test scores on HLM while the MLP scored the best scores on the remaining endpoints. The lowest standard deviation values are achieved by the TabPFN for every endpoint.



(a) Test R2 distributions per model-endpoint combination



(b) Test MAE (log_{10}) distributions per model-endpoint combination

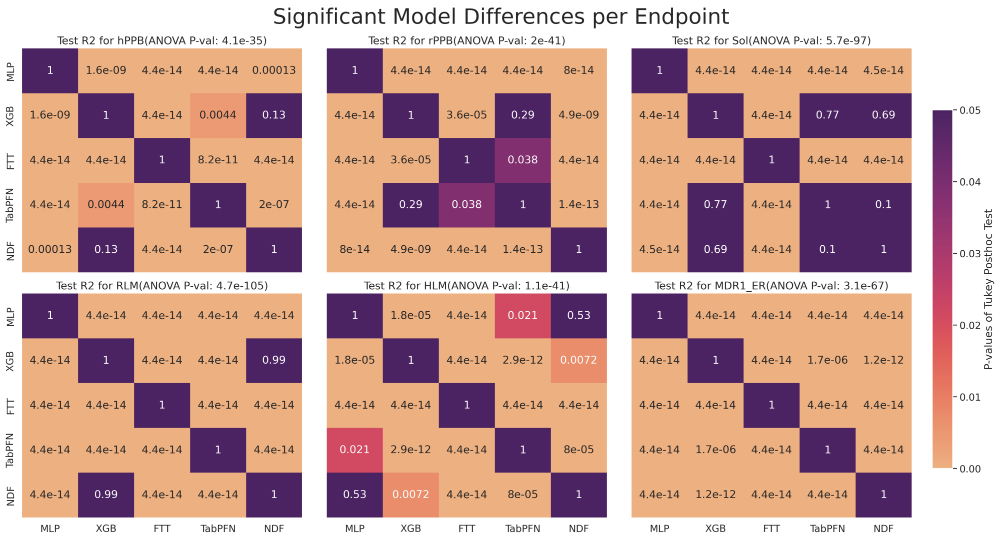
Figure 2: Test performance of the best model architectur per endpoint regarding the R2 and MAE (log_{10}) metric. Each point corresponds to one trained model instance with a different random seed to capture a distribution of the test metrics. The Biogen point is a reference to the best model performance by Fang *et al.*[4] experiment on the public data regarding the R2 score.

Table 2: Testperformance of every model architecture for 30 evaluation runs on R2, MAE and SPR per endpoint capturing mean values and standard deviation

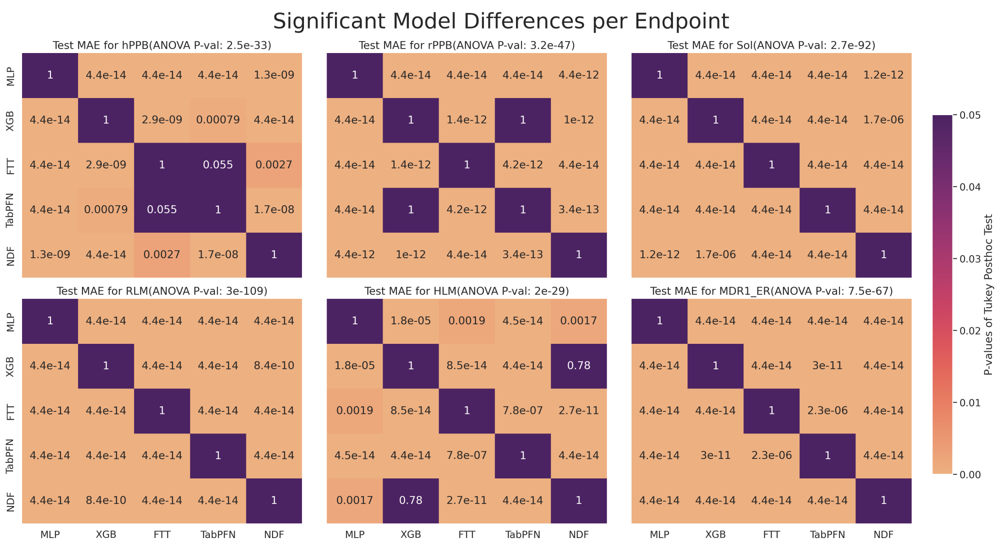
		R2		MAE		SPR	
		mean	std	mean	std	mean	std
Endpoint	Model						
HLM	FTT	0.366196	0.007275	0.382793	0.002901	0.624875	0.005675
	MLP	0.418340	0.024482	0.375901	0.014578	0.636660	0.020122
	NDF	0.408444	0.009587	0.393044	0.004336	0.635272	0.007783
	TabPFN	0.423355	**0.001456**	0.368960	**0.000581**	**0.661207**	**0.001252**
	XGB	**0.434384**	0.005811	**0.366888**	0.002431	0.659852	0.004876
MDR1-ER	FTT	0.514749	0.035335	0.376230	0.018767	0.712664	0.020585
	MLP	**0.678981**	0.004933	**0.297322**	0.002808	**0.796041**	0.003255
	NDF	0.567167	0.009331	0.364007	0.003873	0.753839	0.004674
	TabPFN	0.626152	**0.001196**	0.323651	**0.000577**	0.769841	**0.000867**
	XGB	0.590831	0.005418	0.346865	0.002063	0.756375	0.004104
RLM	FTT	0.448810	0.007435	0.436665	0.003929	0.676999	0.006060
	MLP	**0.521011**	0.006078	**0.400738**	0.003226	**0.725814**	0.004357
	NDF	0.384116	0.010155	0.472673	0.004295	0.638853	0.006458
	TabPFN	0.484334	**0.001054**	0.416583	**0.000434**	0.709351	**0.000801**
	XGB	0.483635	0.004943	0.422276	0.002242	0.703849	0.003633
Sol	FTT	0.013637	0.048945	0.546950	0.019078	0.021830	0.103427
	MLP	**0.407580**	0.015021	**0.380065**	0.012996	**0.558857**	0.026410
	NDF	0.333980	0.011263	0.458386	0.007249	0.447683	0.011949
	TabPFN	0.349094	**0.001385**	0.403110	**0.000474**	0.540974	**0.001161**
	XGB	0.341131	0.005930	0.418536	0.002341	0.520240	0.006020
hPPB	FTT	0.602909	0.006987	0.353853	0.003776	0.777485	0.004346
	MLP	**0.639024**	0.009225	**0.343249**	0.004334	**0.797048**	0.005952
	NDF	0.617922	0.011591	0.356570	0.005785	0.783158	0.008112
	TabPFN	0.629923	**0.002590**	0.350155	**0.001260**	0.792316	**0.001687**
	XGB	0.625117	0.005527	0.360606	0.002543	0.783563	0.003882
rPPB	FTT	0.505550	0.023135	0.409710	0.012451	0.710872	0.015755
	MLP	**0.588841**	0.008544	**0.361912**	0.004699	0.763030	0.005884
	NDF	0.517290	0.023690	0.394077	0.009699	0.725627	0.014269
	TabPFN	0.552698	**0.004114**	0.377532	**0.002063**	0.762246	**0.002733**
	XGB	0.525305	0.008736	0.393678	0.003582	0.742308	0.006177

Significant Results and Effect Size

To measure whether the differences in the test R2 and MAE results (Figure 2) are significant or not we use a pairwise Tukey HSD post hoc test after a repeated measures ANOVA following the suggestions from Ash *et al.*[39] Figure 3 shows the pairwise p-values of the Tukey test for each model pair within each endpoint for both metrics (Figure 3a for R2 and Figure 3b for MAE). The brightness of the heatmap fields indicate the magnitude of the p-value, signifcant results are birghter while unsignificant results are colored dark purple. Overall most of the results are significant differently from each other yielding some exceptions for certain model combinations in some of the endpoints. Especially the supperior performance of the MLP seems significant for both metrics on every endpoint except HLM, the only endpoint where the MLP did not achieved the best test scores.



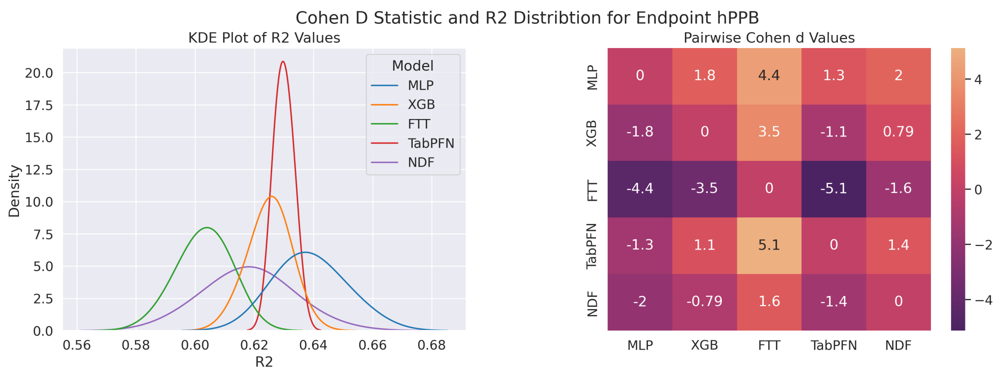
(a) Pvalues to identify significant differences in the R2 distributions per model-endpoint combination



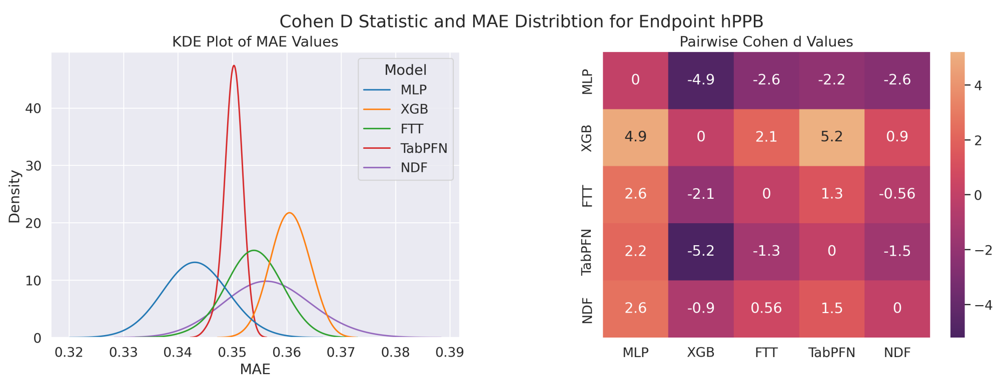
(b) Pvalues to identify significant differences in the MAE (log_{10}) distributions per model-endpoint combination

Figure 3: Heatmap of p-values calculated by a tukey HSD posthoc test after a significant multi measurement ANOVA. An orange color shows if the R2 distributions of one model is significant differently from another model. This is done for each endpoint-model combination.

We use the Cohen D[40] to measure the pairwise effect size between two test distributions. Figure 4 shows the R2 test distributions smoothed with a KDE for endpoint hPPB (see Supplement for the other endpoints). The heatmap shows the pairwise Cohen D values where higher values indicating a larger effect size. This quantifies the differences from Figure 3 as standardized distances between mean values of the test R2 and MAE distributions. The highest effect size can be observed for the MLP on endpoint hPPB for both metrics (highest Cohen D value for R2 and lowest for MAE) since every pairwise comparison is positive for the R2 differences and negative for the MAE differences.



(a) Visualized and quantified effect size based on the R2 distributions for various model on the hppb testset.



(b) Visualized and quantified effect size based on the MAE (log_{10}) distributions for various model on the hppb testset.

Figure 4: KDE for the R2 and MAE test performance for MLP, XGB, FTT, TabPFN and NDF. A heatmap shows the pairwise Cohen D scores.

Applicability Domain

Figure 5 shows the relationship between the prediction error and the structural similarity for every model architecture on the endpoint hPPB. The error rate gradually increases for test compounds that have a higher similarity regarding the training compounds showing that ML models are affected by similarities between training and test compounds.[42,43]

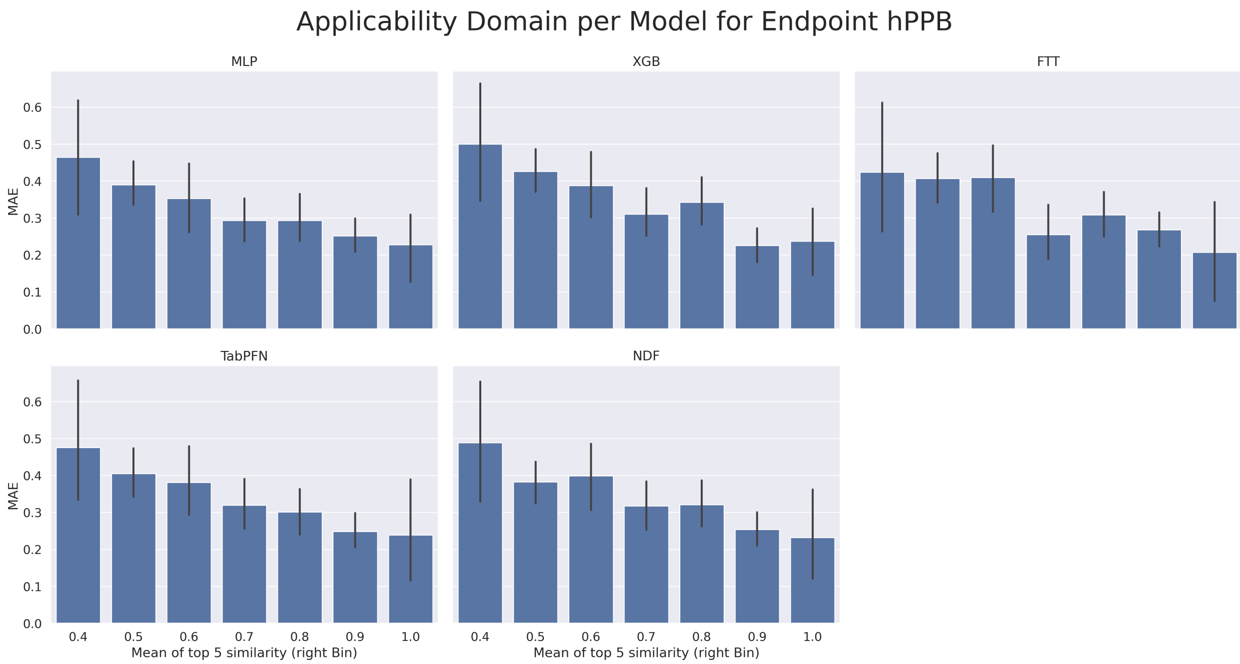


Figure 5: The Applicability domain on hPPB for each model.

MLP shows a better generalization ability for the endpoints rPPB and RLM (see Supplement) since the correlation between similarity and the MAE is not as strong compared to the other models.

Correlation between Performance Metrics and Datapoints

Figure 6 shows the correlation by Pearson between the test R2/MAE and the amount of datapoints per endpoint. R2 and MAE show a negative linear relationship indicating a higher R2 when the MAE is lower. A negative correlation is also present between R2 and the amount of data: The test R2 is higher when the dataset is smaller and the other way around. A linear relationship between MAE and the amount of data is not present.

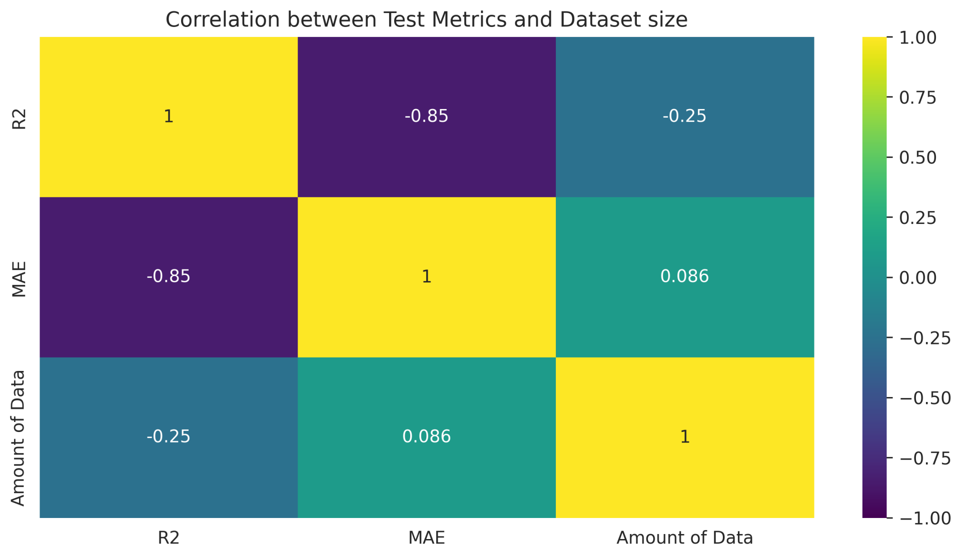


Figure 6: Pearson R between amount of Datapoints per Endpoint and Test R2/MAE aggregated by the mean

Discussion

Table 2 shows that not a single one of the modern tabular approaches (FTT, NDF, TabPFN) achieved the best scores in one endpoint since it is always a MLP/XGB that achieved supperior performance. This result remains significant regarding the top result defined by the highest test R2 and lowest test MAE value. Also, the applicability domain shows the same correlation trend suggesting that every model performs better on compounds which are more similar to seen compounds indicating a structural similarity bias (see Supplement and Figure 5). The MLP shows a less correlated relationship between compound similarity and MAE indicating better generalization ability. This could be further enhanced through a Multitask Learning strategy[44] which is not possible for XGB models since they are built for singletask problems.

The FTT shows a collapse for endpoint Sol with a mean R2 of 0.014 and mean MAE of 0.547. Besides this, we needed to project the features on a lower dimension since a nvidia A100 with 80Gb Vram was not enough for the full feature space since the FTT scales very poorly with more features.[18] Experiments with the full feature space or lower feature space would bring more insight for the application of FTT for ADME prediction. Nevertheless, it should be noted that the FTT wasnt designed for the chemoinformatic domain since no categorical embeddings are used and the feature spaces are much higher compared to other domains.

The performance gap between XGB and NDF underlines the popularity of XGB for compound related tasks.[4,14] Possible reasons for this behaviour could be a difference between bagging vs boosting or hard splits vs soft splits related to high dimensional tabular data for chemoinformatical tasks.

However, this does not mean that TabPFN isnt suitable for ADME prediction or any other tabular based chemoinformatic task because the difference between the highest and the remaining mean values in Table 2 is not very high. Table 2 also shows that TabPFN yields the smallest standard deviation for every endpoint-metric combination. Hollmann *et al.*[20] claim that TabPFN needs no hyperparameter tuning which is why TabPFN does provide only a small hyperparameter space leading to low variance when retrained with different random seeds. The authors advise that only small tabular datasets with ≤ 1000 samples and ≤ 100 features should be used for training a TabPFN model. We violate against both suggestions since every endpoint has more than 1000 samples (except rPPB) and more than 100 features since very SMILES string is featurized in the same way with 1340 features. The TabPFN could potentially reach even better results when the pretraining is done one chemoinformatic datasets since the TabPFN was pretrained with synthetical data from various domains. Anyhow, the TabPFN achieves the second best results for the test metrics on every endpoint making it a promising model choice for smaller datasets with fewer features.

We also observe that the test R2 is correlated with the amount of datapoints per endpoint (Figure 6) indicating that endpoints with more compounds are more difficult to predict.

Conclusion

Our results demonstrate that while modern deep learning architectures perform competitively, they do not consistently outperform traditional ML models for tabular based ADME prediction. XGBoost and MLPs remain a highly effective choice achieving superior results across multiple endpoints. These findings suggest that, despite the growing interest in Transformer based models, traditional machine learning techniques like MLPs and XGB remain the most practical and reliable choice for structured ADME prediction tasks. However, TabPFN proved itself as a promising contender since it achieved the second-best test results on every endpoint with the lowest standard deviation although the model assumption regarding the shape of the dataset are violated.

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